

RETAIL SALES ANALYTICS

Sales Performance & Business Insights

Data Source: Sample Superstore Dataset
Analysis Tools: Python, Pandas, PostgreSQL / SQL, Power BI
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Total Sales	Total Profit	Orders	Quantity	Avg. Discount
\$2.30M	\$0.29M	9,994	37,873	15.62%

This report presents an end-to-end retail analytics project covering data preparation, exploratory analysis, SQL-based business analysis, and Power BI visualisation.

1. Project Overview

The Retail Sales Analytics project analyses the Sample Superstore dataset to understand sales performance, profitability, customer segments, regional performance, product categories, shipping behaviour, and discount impact. The project combines Python-based exploratory analysis, PostgreSQL/SQL analysis, and an interactive Power BI dashboard.

Business Objectives

- Measure overall sales, profit, order volume, quantity and discount performance.
- Identify high-performing categories, sub-categories, regions, states and customer segments.
- Investigate profitability and identify areas generating losses.
- Use SQL to answer operational and business questions.
- Present decision-ready insights through Power BI.

Dataset Profile

Records	9,994
Columns	13
Categories	3
Sub-Categories	17
Regions	4
Customer Segments	3
States	49
Ship Modes	4

2. Data Preparation & Methodology

Python / Exploratory Data Analysis

- Loaded the Sample Superstore CSV using Pandas.
- Inspected shape, data types, missing values, duplicates and descriptive statistics.
- Removed duplicate records in the notebook workflow.
- Reviewed correlations among numeric variables.
- Analysed sales and profit by category, region, segment, state, city and sub-category.
- Examined discount, sales and profit distributions and the relationship between sales and profit.

SQL Analysis

- Created a PostgreSQL table named **retail_sales**.
- Performed basic queries for totals, filtering, sorting and loss-making orders.
- Used GROUP BY and HAVING for category, region and segment analysis.
- Used CASE expressions to classify sales and profit status.
- Used RANK, DENSE_RANK and ROW_NUMBER window functions.
- Used CTEs and subqueries for advanced analysis.
- Calculated category-level profit margins.

Power BI

- Built an interactive dashboard for sales performance and business insights.
- Included slicers for Region, Category, Segment and Ship Mode.
- Presented KPIs for sales, profit, orders, quantity and average discount.

3. Sales & Profitability Performance

The dataset generated **\$2,297,200.86** in total sales and **\$286,397.02** in total profit, producing an overall profit margin of **12.47%**. The average sales value per record was **\$229.86** and the average discount was **15.62%**.

Performance by Category

Category	Sales	Profit	Quantity	Orders	Margin
Technology	\$836,154.03	\$145,454.95	6,939	1,847	17.40%
Furniture	\$741,999.80	\$18,451.27	8,028	2,121	2.49%
Office Supplies	\$719,047.03	\$122,490.80	22,906	6,026	17.04%

- **Technology** generated the highest category sales at approximately \$836K and the highest category profit at approximately \$145K.
- **Office Supplies** generated approximately \$719K in sales but delivered about \$122K profit, giving it a strong margin.
- **Furniture** generated approximately \$742K in sales but only about \$18K profit, making it the weakest category by profit margin.

4. Regional & Customer Segment Analysis

Regional Performance

Region	Sales	Profit	Orders
West	\$725,457.82	\$108,418.45	3,203
East	\$678,781.24	\$91,522.78	2,848
Central	\$501,239.89	\$39,706.36	2,323
South	\$391,721.91	\$46,749.43	1,620

- **West** was the strongest region by both sales (\$725K) and profit (\$108K).
- **South** had the lowest sales (\$392K) but remained profitable at about \$47K.

Customer Segment Performance

Segment	Sales	Profit	Orders
Consumer	\$1,161,401.34	\$134,119.21	5,191
Corporate	\$706,146.37	\$91,979.13	3,020
Home Office	\$429,653.15	\$60,298.68	1,783

- **Consumer** was the largest segment, contributing about \$1.16M in sales and \$134K profit.
- Corporate generated about \$706K in sales and \$92K profit, while Home Office generated about \$430K in sales and \$60K profit.

5. Product, State & Discount Insights

Top Sub-Categories by Sales

Sub-Category	Sales	Profit
Phones	\$330,007.05	\$44,515.73
Chairs	\$328,449.10	\$26,590.17
Storage	\$223,843.61	\$21,278.83
Tables	\$206,965.53	\$-17,725.48
Binders	\$203,412.73	\$30,221.76
Machines	\$189,238.63	\$3,384.76
Accessories	\$167,380.32	\$41,936.64
Copiers	\$149,528.03	\$55,617.82
Bookcases	\$114,880.00	\$-3,472.56
Appliances	\$107,532.16	\$18,138.01

- **Phones** and **Chairs** were the top two sub-categories by sales, each contributing roughly \$330K.
- **Tables** produced approximately \$207K in sales but a loss of about \$17.7K.
- **Bookcases** and **Supplies** also recorded negative total profit, indicating areas requiring margin and pricing review.

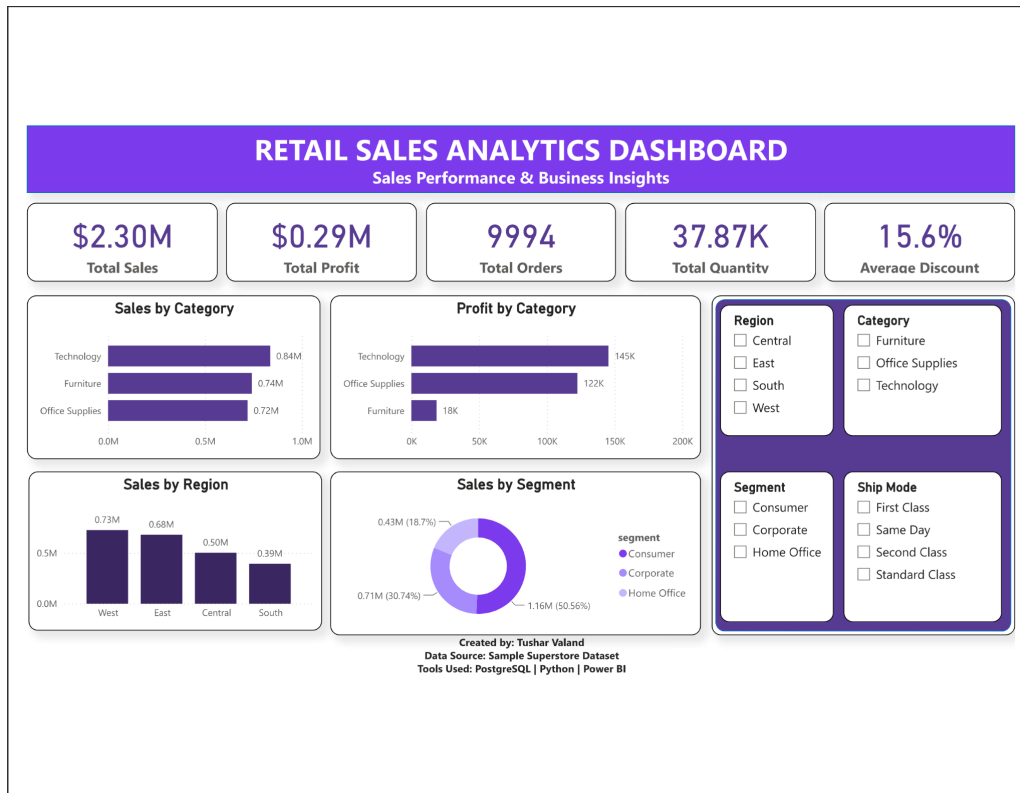
Top States by Sales

Rank	State	Sales
1	California	\$457,687.63
2	New York	\$310,876.27
3	Texas	\$170,188.05
4	Washington	\$138,641.27
5	Pennsylvania	\$116,511.91
6	Florida	\$89,473.71
7	Illinois	\$80,166.10
8	Ohio	\$78,258.14
9	Michigan	\$76,269.61
10	Virginia	\$70,636.72

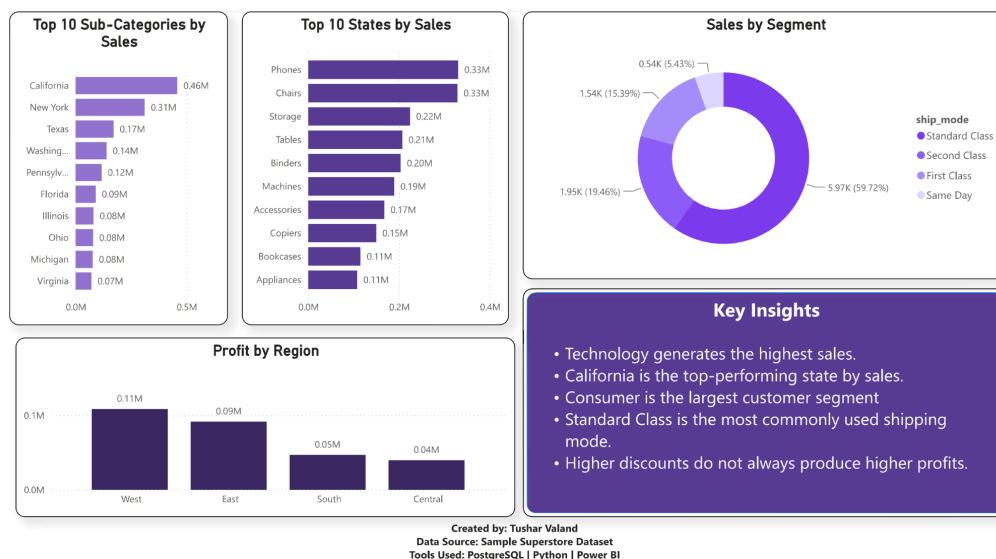
- **California** was the top state by sales at approximately \$458K, followed by New York at approximately \$311K.
- The dataset contains **1,871 loss-making records**, with combined losses of approximately **\$156,131**.

6. Power BI Dashboard

The Power BI dashboard consolidates the analysis into an interactive management view. It includes KPI cards, category and regional comparisons, segment analysis, and slicers for Region, Category, Segment and Ship Mode.



Dashboard Page 1 — Sales Performance



Dashboard Page 2 — Detailed Insights

7. Key Insights & Business Recommendations

Key Insights

- Technology is the strongest category by both sales and profit.
- West is the leading region by sales and profit.
- Consumer is the largest customer segment and contributes the highest sales and profit.
- Standard Class is the most frequently used shipping mode and accounts for the largest sales share.
- California is the top-performing state by sales.
- Higher discounts do not automatically translate into higher profits.
- Tables, Bookcases and Supplies are notable loss-making sub-categories and require closer review.

Business Recommendations

- **Review loss-making products:** Investigate pricing, discount levels, product costs and shipping costs for Tables, Bookcases and Supplies.
- **Protect profitable Technology sales:** Continue investment in high-performing technology products while monitoring margin.
- **Strengthen the South region:** Investigate lower sales volume and identify category or customer opportunities for growth.
- **Optimise discount strategy:** Use targeted discounts rather than broad discounting, especially where margins are already weak.
- **Focus on Consumer customers:** Develop retention, cross-selling and personalised campaigns for the largest segment.
- **Use regional and state insights:** Prioritise high-performing markets while testing targeted growth strategies in underperforming areas.

8. Conclusion

The Retail Sales Analytics project demonstrates an end-to-end approach to turning transactional data into business insight. Python was used for data inspection and exploratory analysis, PostgreSQL/SQL was used to answer structured business questions, and Power BI was used to communicate the results through an interactive dashboard.

The analysis shows a business generating approximately **\$2.30M in sales** and **\$0.29M in profit**, with Technology, the West region and Consumer segment emerging as major performance drivers. At the same time, loss-making sub-categories demonstrate the importance of looking beyond revenue and monitoring profitability.

Skills Demonstrated

- Python and Pandas
- Exploratory Data Analysis (EDA)
- SQL and PostgreSQL
- Data Cleaning and Quality Checks
- Aggregations, CASE expressions, CTEs and Window Functions
- Power BI Dashboard Development
- Data Visualisation
- Business Analysis and Insight Generation
- Data-Driven Recommendations

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